

Practice Midterm 1 Solutions

Math 124, Fall 2026

Name: _____

username: _____

UMID: _____

Room: ☐ B844 East Hall

Instructions

- This exam consists of 7 problems across 10 pages, worth a total of 100 points.
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page where a space is provided.
- For free response problems, **you must show all of your work**, and write your final answer in the small box when one is provided.
- For multiple choice problems, completely fill in bubbles and square boxes.
 - ☐ A bubble means that you should only select one choice.
 - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 4×6 -inch index card**, handwritten directly on the card. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

By signing below, I affirm that this work is my own and that I have followed the exam's resource and collaboration rules.

This page has been intentionally left blank.

Problem 1 (9 pts)

Consider the points

$$P = (-7, 8, -6), \quad Q = (8, -12, 19)$$

- a) (2 pts) Let $\vec{d}$ be the vector pointing from P to Q . Find $\vec{d}$. Give your answer as a vector.

Solution: Subtract the coordinates of the starting point from those of the ending point.

$$\vec{d} = \begin{bmatrix} 8 - (-7) \\ -12 - 8 \\ 19 - (-6) \end{bmatrix} = \begin{bmatrix} 15 \\ -20 \\ 25 \end{bmatrix}$$

- b) (3 pts) Find the distance between P and Q .

Solution: The distance is the length of the vector from P to Q . Notice that

$$\vec{d} = 5 \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$$

The length of a scalar multiple is the absolute value of the scalar times the original length. So,

$$\|\vec{d}\| = 5 \left\| \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix} \right\| = 5\sqrt{3^2 + (-4)^2 + 5^2} = 5\sqrt{50} = \boxed{25\sqrt{2}}$$

- c) (4 pts) Find the coordinates of the point R that is $\frac{3}{5}$ of the way from P to Q .

Solution: Start at P and add $\frac{3}{5}$ of the vector pointing from P to Q .

$$\begin{bmatrix} -7 \\ 8 \\ -6 \end{bmatrix} + \frac{3}{5} \begin{bmatrix} 15 \\ -20 \\ 25 \end{bmatrix} = \begin{bmatrix} -7 \\ 8 \\ -6 \end{bmatrix} + \begin{bmatrix} 9 \\ -12 \\ 15 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 9 \end{bmatrix}$$

So, $\boxed{R = (2, -4, 9)}$.

Problem 2 (16 pts)

Suppose $\vec{u}, \vec{v} \in \mathbb{R}^3$ satisfy

$$\|\vec{u}\| = 5, \quad \|\vec{v}\| = 4, \quad \vec{u} \cdot \vec{v} = -6$$

a) (7 pts)

I) (5 pts) Find $(5\vec{u} - 6\vec{v}) \cdot (8\vec{u} + 10\vec{v})$. Show your work.

Solution: Distribute the dot product, using $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$ and $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$.

$$\begin{aligned} (5\vec{u} - 6\vec{v}) \cdot (8\vec{u} + 10\vec{v}) &= 40\|\vec{u}\|^2 + (50 - 48)(\vec{u} \cdot \vec{v}) - 60\|\vec{v}\|^2 \\ &= 20(2\|\vec{u}\|^2 - 3\|\vec{v}\|^2) + 2(\vec{u} \cdot \vec{v}) \\ &= 20(50 - 48) + 2(-6) = 40 - 12 = \boxed{28} \end{aligned}$$

II) (2 pts) What type of angle is formed by $5\vec{u} - 6\vec{v}$ and $8\vec{u} + 10\vec{v}$? Select one.

☒ Acute ☐ Right ☐ Obtuse

Solution: The dot product is positive, so the angle is acute. Recall that $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\theta$; a positive dot product means $\cos\theta > 0$.

b) (5 pts) Find $\|\vec{u} - \vec{v}\|$.

Solution: First find the squared length by taking the dot product of the vector with itself.

$$\begin{aligned} \|\vec{u} - \vec{v}\|^2 &= (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) \\ &= \|\vec{u}\|^2 - 2(\vec{u} \cdot \vec{v}) + \|\vec{v}\|^2 \\ &= 25 - 2(-6) + 16 = 53 \end{aligned}$$

Therefore, $\|\vec{u} - \vec{v}\| = \sqrt{53}$.

c) (4 pts) Could a unit vector $\vec{w} \in \mathbb{R}^3$ satisfy $\vec{u} \cdot \vec{w} = 9$? Explain why or why not.

Solution: No. By the Cauchy-Schwarz inequality, a unit vector $\vec{w}$ must satisfy

$$|\vec{u} \cdot \vec{w}| \leq \|\vec{u}\|\|\vec{w}\| = 5(1) = 5$$

Since $9 > 5$, the proposed dot product is impossible.

Problem 3 (14 pts)

- a) (3 pts) Let ℓ be the line in $\mathbb{R}^2$ defined by $5x + 7y = 35$. Find nonzero vectors $\vec{n}$ and $\vec{d}$ such that $\vec{n}$ is perpendicular to ℓ and $\vec{d}$ is parallel to ℓ .

Solution: The coefficients of x and y give a normal vector. A direction vector must be orthogonal to it, so one choice is

$$\vec{n} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}, \quad \vec{d} = \begin{bmatrix} 7 \\ -5 \end{bmatrix}$$

These work because $\vec{n} \cdot \vec{d} = 5(7) + 7(-5) = 0$. Any nonzero scalar multiples also work.

- b) (4 pts) Write the line ℓ , defined by $5x + 7y = 35$, in vector-parametric form.

Solution: The point $(7, 0)$ is on ℓ , and part (a) gives a direction vector. So, one vector-parametric form is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix} + t \begin{bmatrix} 7 \\ -5 \end{bmatrix}, \quad t \in \mathbb{R}$$

Substituting gives $5(7 + 7t) + 7(-5t) = 35$ for every t .

- c) (7 pts) Consider another line ℓ' , defined by $7x - 5y = 13$. Which of the following are valid scalar-parametric forms of ℓ' ? **Select all that apply.** In each option, $t \in \mathbb{R}$.

- ☒ 1. $\begin{cases} x = 9 + 5t \\ y = 10 + 7t \end{cases}$
☐ 2. $\begin{cases} x = 9 + 7t \\ y = 10 - 5t \end{cases}$
☒ 3. $\begin{cases} x = -6 - 10t \\ y = -11 - 14t \end{cases}$

- ☒ 4. $\begin{cases} x = 14 + 15t \\ y = 17 + 21t \end{cases}$
☐ 5. $\begin{cases} x = 8 + 5t \\ y = 9 + 7t \end{cases}$

Solution: Select the first, third, and fourth options. Substitute each option into $7x - 5y = 13$:

- **Option 1:** $x = 9 + 5t$, $y = 10 + 7t$. Substituting gives $7(9 + 5t) - 5(10 + 7t) = 63 + 35t - 50 - 35t = 13$. The equation is satisfied for every t , so this option is valid.
- **Option 2:** $x = 9 + 7t$, $y = 10 - 5t$. Substituting gives $7(9 + 7t) - 5(10 - 5t) = 63 + 49t - 50 + 25t = 13 + 74t$. The equation is satisfied only when $t = 0$, so this option is invalid.
- **Option 3:** $x = -6 - 10t$, $y = -11 - 14t$. Substituting gives $7(-6 - 10t) - 5(-11 - 14t) = -42 - 70t + 55 + 70t = 13$. The equation is satisfied for every t , so this option is valid.
- **Option 4:** $x = 14 + 15t$, $y = 17 + 21t$. Substituting gives $7(14 + 15t) - 5(17 + 21t) = 98 + 105t - 85 - 105t = 13$. The equation is satisfied for every t , so this option is valid.
- **Option 5:** $x = 8 + 5t$, $y = 9 + 7t$. Substituting gives $7(8 + 5t) - 5(9 + 7t) = 56 + 35t - 45 - 35t = 11 \neq 13$. The equation is never satisfied, so this option is invalid.

Problem 4 (14 pts)

The vectors

$$\vec{u}_1 = \frac{1}{19} \begin{bmatrix} -15 \\ 10 \\ 6 \end{bmatrix}, \quad \vec{u}_2 = \frac{1}{19} \begin{bmatrix} -6 \\ -15 \\ 10 \end{bmatrix}, \quad \vec{u}_3 = \frac{1}{19} \begin{bmatrix} 10 \\ 6 \\ 15 \end{bmatrix}$$

form an orthonormal basis of $\mathbb{R}^3$.

- a) (2 pts) Given that these vectors form an orthonormal basis of $\mathbb{R}^3$, find the following quantities.

$$\vec{u}_1 \cdot \vec{u}_3 = \boxed{0} \quad \|\vec{u}_1\| = \boxed{1}$$

Solution: Orthonormal means that distinct basis vectors are orthogonal and each basis vector has length 1. Therefore,

$$\boxed{\vec{u}_1 \cdot \vec{u}_3 = 0}, \quad \boxed{\|\vec{u}_1\| = 1}$$

- b) (4 pts) Find $\|5\vec{u}_1 - 7\vec{u}_2\|$.

Solution: Since $\vec{u}_1$ and $\vec{u}_2$ are orthogonal unit vectors,

$$\begin{aligned} \|5\vec{u}_1 - 7\vec{u}_2\|^2 &= 25\|\vec{u}_1\|^2 - 70(\vec{u}_1 \cdot \vec{u}_2) + 49\|\vec{u}_2\|^2 \\ &= 25 + 49 = 74 \end{aligned}$$

So, $\boxed{\|5\vec{u}_1 - 7\vec{u}_2\| = \sqrt{74}}.$

- c) (8 pts) Let $\vec{w} = \begin{bmatrix} -4 \\ 3 \\ 4 \end{bmatrix}$. Write $\vec{w}$ as a linear combination of $\vec{u}_1$, $\vec{u}_2$, and $\vec{u}_3$.

Solution: For an orthonormal basis, the coefficient of $\vec{u}_i$ is $\vec{w} \cdot \vec{u}_i$. Taking dot products gives

$$\begin{aligned} \vec{w} \cdot \vec{u}_1 &= \frac{(-4)(-15) + 3(10) + 4(6)}{19} = \frac{114}{19} = 6 \\ \vec{w} \cdot \vec{u}_2 &= \frac{(-4)(-6) + 3(-15) + 4(10)}{19} = \frac{19}{19} = 1 \\ \vec{w} \cdot \vec{u}_3 &= \frac{(-4)(10) + 3(6) + 4(15)}{19} = \frac{38}{19} = 2 \end{aligned}$$

Therefore,

$$\boxed{\vec{w} = 6\vec{u}_1 + \vec{u}_2 + 2\vec{u}_3}$$

Problem 5 (19 pts)

Let P be the plane through the points

$$A = (8, -15, 6), \quad B = (13, -5, 11), \quad C = (15, -8, 7)$$

- a) (5 pts) Express P in vector-parametric form.

Solution: Use A as a starting point. The directions $\vec{d}_1$ from A to B and $\vec{d}_2$ from A to C are not scalar multiples, so they span the plane.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -15 \\ 6 \end{bmatrix} + s \underbrace{\begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}}_{\vec{d}_1} + t \underbrace{\begin{bmatrix} 7 \\ 7 \\ 1 \end{bmatrix}}_{\vec{d}_2}, \quad s, t \in \mathbb{R}$$

- b) (8 pts) Write an equation for P in the form $ax + by + cz = d$.

Solution: The coefficients a, b, c of a nonzero normal vector must satisfy

$$\begin{cases} 5a + 10b + 5c = 0 \\ 7a + 7b + c = 0 \end{cases}$$

The second equation gives $c = -7a - 7b$. Substituting into the first gives $-30a - 25b = 0$, or $6a + 5b = 0$.

Choose $a = 5$. Then $b = -6$ and $c = -7(5) - 7(-6) = 7$. Since A lies on the plane $5x - 6y + 7z = d$,

$$d = 5(8) - 6(-15) + 7(6) = 172$$

So, the equation is $\boxed{5x - 6y + 7z = 172}$.

- c) (6 pts) Suppose C is replaced by $C' = (23, 15, 21)$, while A and B stay the same. Do the three points A, B, C' determine a unique plane? Explain what changes geometrically.

Solution: $\boxed{\text{No}}$. The new direction vector is

$$\vec{d}_3 = \begin{bmatrix} 23 - 8 \\ 15 - (-15) \\ 21 - 6 \end{bmatrix} = \begin{bmatrix} 15 \\ 30 \\ 15 \end{bmatrix} = 3\vec{d}_1$$

Unlike $\vec{d}_1$ and $\vec{d}_2$, the vectors $\vec{d}_1$ and $\vec{d}_3$ are linearly dependent. The three points are now collinear, so infinitely many planes contain their line.

Problem 6 (18 pts)

Consider the line ℓ in $\mathbb{R}^3$ given by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -8 \\ 11 \end{bmatrix} + t \begin{bmatrix} 5 \\ -6 \\ 7 \end{bmatrix}, \quad t \in \mathbb{R}$$

- a) (10 pts) Find equations for two distinct planes whose intersection is exactly ℓ . Your equations should both be of the form $ax + by + cz = d$.

Solution: Each plane must contain $(7, -8, 11)$, and its normal must be orthogonal to the line's direction $\begin{bmatrix} 5 \\ -6 \\ 7 \end{bmatrix}$. Two nonparallel choices are

$$\vec{n}_1 = \begin{bmatrix} 6 \\ 5 \\ 0 \end{bmatrix}, \quad \vec{n}_2 = \begin{bmatrix} 7 \\ 0 \\ -5 \end{bmatrix}$$

Their dot products with the direction vector are $30 - 30 = 0$ and $35 - 35 = 0$. Substituting the point into each plane gives

$$6(7) + 5(-8) = 2, \quad 7(7) - 5(11) = -6$$

So, one answer is

$$6x + 5y + 0z = 2,$$

$$7x + 0y - 5z = -6$$

- b) (8 pts) A student proposes the equations $11x + 15y + 5z = 12$ and $55x + 75y + 25z = 60$. Every point on ℓ satisfies both. Explain why these equations, unlike those in part (a), do not intersect exactly at the line ℓ . Describe their intersection and give one point in it that is not on ℓ .

Solution: The second equation is five times the first, so they describe the same plane. Their intersection is that entire plane, which contains points outside ℓ .

One such point is $(2, 1, -5)$, since

$$11(2) + 15(1) + 5(-5) = 12, \quad 55(2) + 75(1) + 25(-5) = 60$$

It is not on ℓ : its x -coordinate would require $7 + 5t = 2$, or $t = -1$, but then the line's y -coordinate would be $-8 - 6(-1) = -2$, rather than 1.

Problem 7 (10 pts)

Let

$$\vec{v} = \begin{bmatrix} 6 \\ 8 \\ -8 \end{bmatrix}, \quad \ell = \text{span} \left(\begin{bmatrix} 5 \\ -5 \\ 7 \end{bmatrix} \right), \quad P : 5x - 5y + 7z = 0$$

- a) (6 pts) Find the orthogonal projection of $\vec{v}$ onto ℓ .

Solution: Let $\vec{w} = \begin{bmatrix} 5 \\ -5 \\ 7 \end{bmatrix}$. The projection onto its span is

$$\text{proj}_{\ell}(\vec{v}) = \frac{\vec{v} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w}$$

Here,

$$\vec{v} \cdot \vec{w} = 6(5) + 8(-5) + (-8)(7) = -66, \quad \vec{w} \cdot \vec{w} = 25 + 25 + 49 = 99$$

Therefore,

$$\text{proj}_{\ell}(\vec{v}) = -\frac{2}{3} \begin{bmatrix} 5 \\ -5 \\ 7 \end{bmatrix} = \boxed{\begin{bmatrix} -10/3 \\ 10/3 \\ -14/3 \end{bmatrix}}$$

- b) (4 pts) Find the orthogonal projection of $\vec{v}$ onto P .

Solution: The plane passes through the origin and has normal $\vec{w}$, so it is perpendicular to ℓ . To project onto the plane, subtract the component along the normal.

$$\text{proj}_P(\vec{v}) = \vec{v} - \text{proj}_{\ell}(\vec{v}) = \begin{bmatrix} 6 \\ 8 \\ -8 \end{bmatrix} - \begin{bmatrix} -10/3 \\ 10/3 \\ -14/3 \end{bmatrix} = \boxed{\begin{bmatrix} 28/3 \\ 14/3 \\ -10/3 \end{bmatrix}}$$

This vector lies in P , since $5(28/3) - 5(14/3) + 7(-10/3) = 0$, and the subtracted component is perpendicular to P .